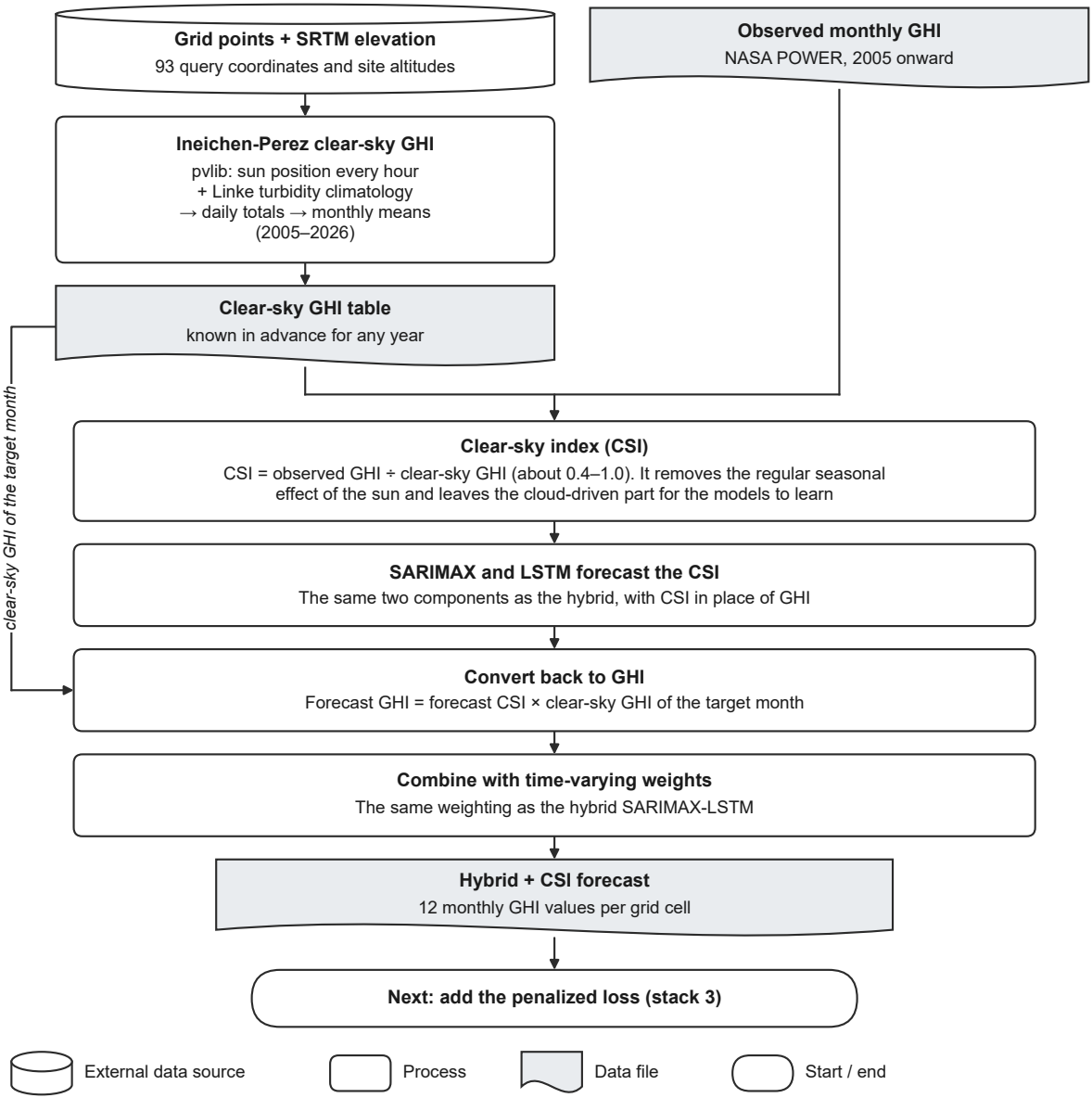

Grid points + SRTM elevation
93 query coordinates and site altitudes
Observed monthly GHI
NASA POWER, 2005 onward
Ineichen-Perez clear-sky GHI
pvlib: sun position every hour
+ Linke turbidity climatology
→ daily totals → monthly means
(2005–2026)
Clear-sky GHI table
known in advance for any year
clear-sky GHI of the target month
Clear-sky index (CSI)
CSI = observed GHI ÷ clear-sky GHI (about 0.4–1.0). It removes the regular seasonal effect of the sun and leaves the cloud-driven part for the models to learn
SARIMAX and LSTM forecast the CSI
The same two components as the hybrid, with CSI in place of GHI
Convert back to GHI
Forecast GHI = forecast CSI × clear-sky GHI of the target month
Combine with time-varying weights
The same weighting as the hybrid SARIMAX-LSTM
Hybrid + CSI forecast
12 monthly GHI values per grid cell
Next: add the penalized loss (stack 3)
External data source
Process
Data file
Start / end